

BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number: _BIT/2024/73846

School/Faculty: _School of Computing and Informatics

Programme: _BIT

Academic Year: __2026

Semester: __Year 3 Sem 2

Lecturer: __Mr. Nyoro

Phone Number: _0758989285

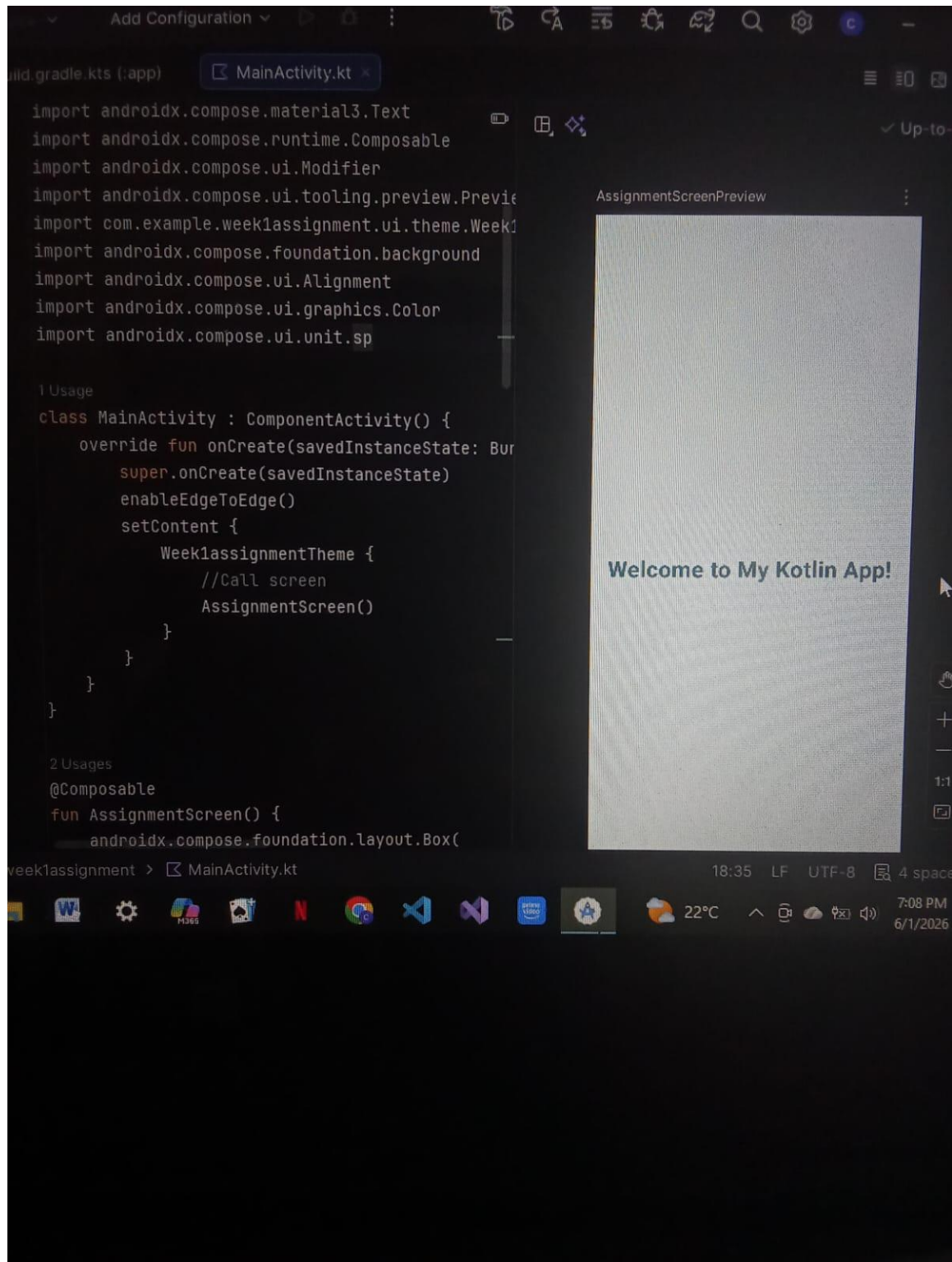
Instructions

This logbook is used to document weekly learning activities, practical exercises, skills acquired, challenges encountered, solutions applied, and reflections. Students should complete all sections every week.

Each week to follow remarks in week 1 section.

Autogenerated Table of content

Week 1: Introduction to Mobile Application Development



Practical Activities Undertaken

only screenshots needed and labeling

Screenshots/Evidence Attached (Yes/No)

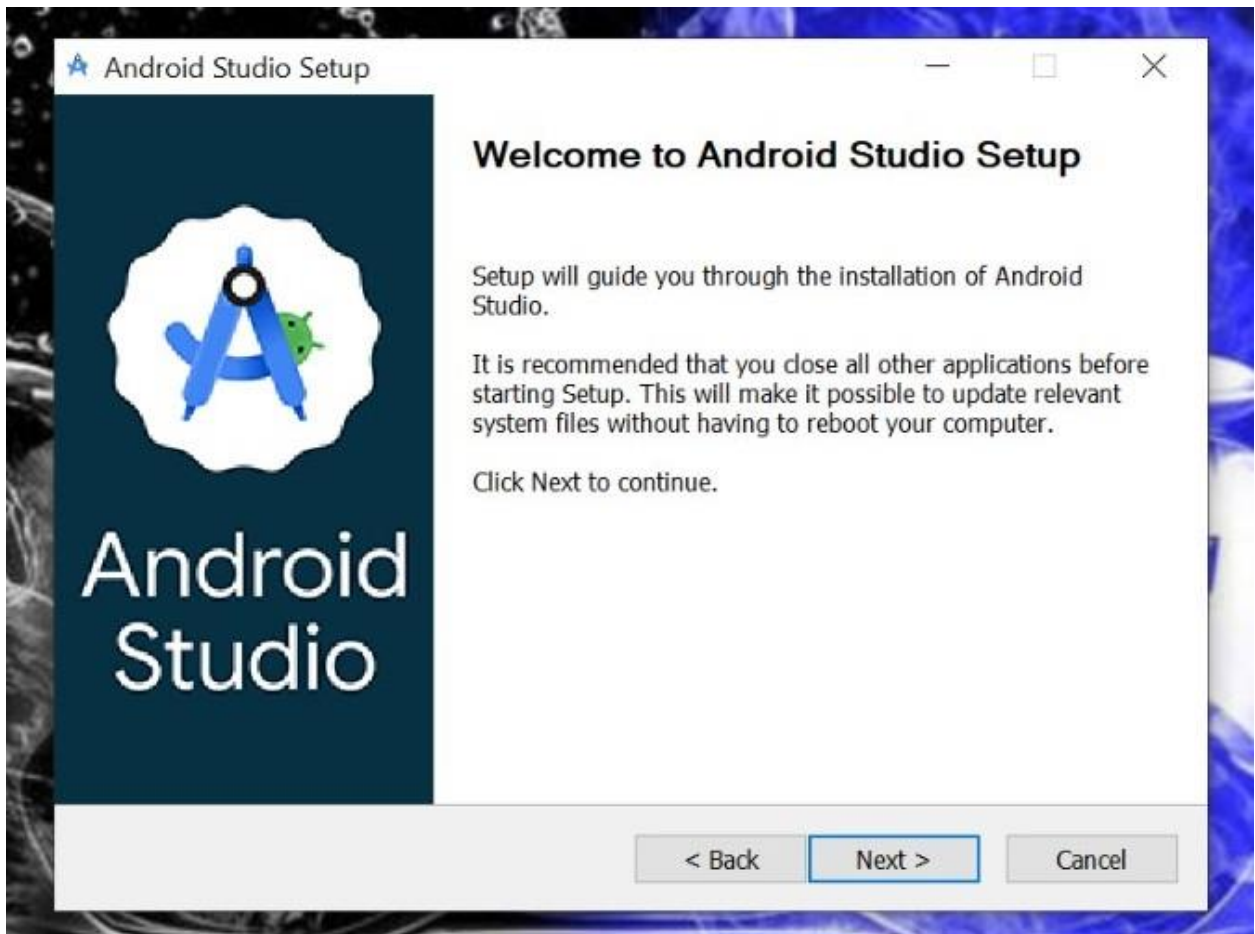
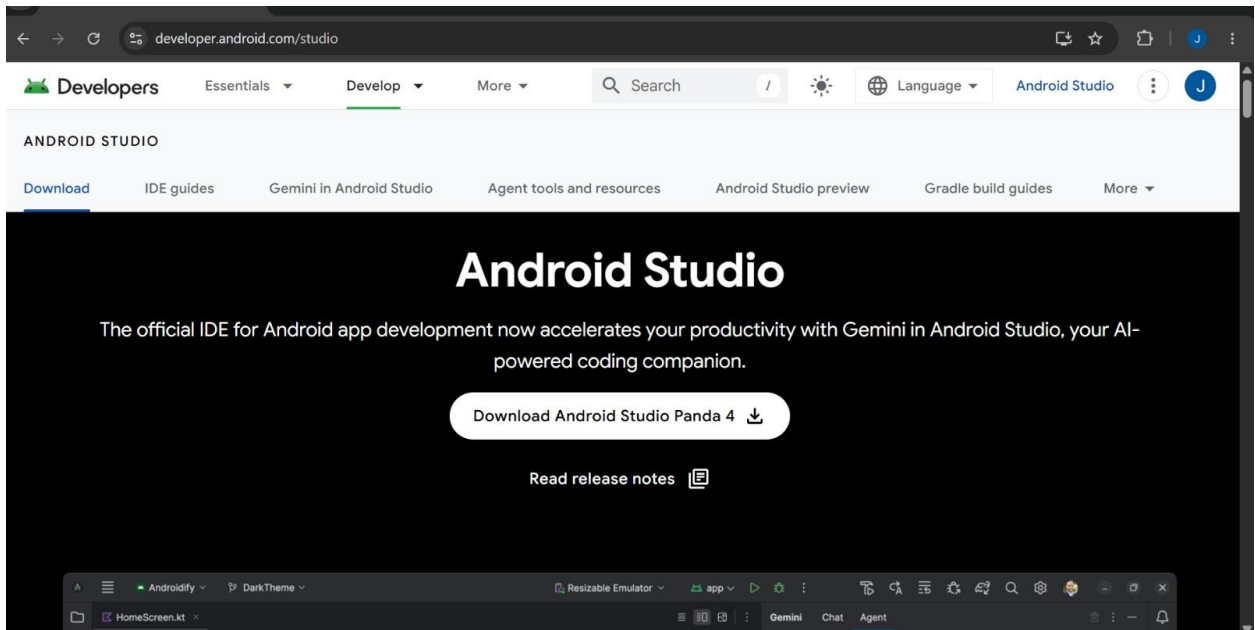
Tools/Software Used

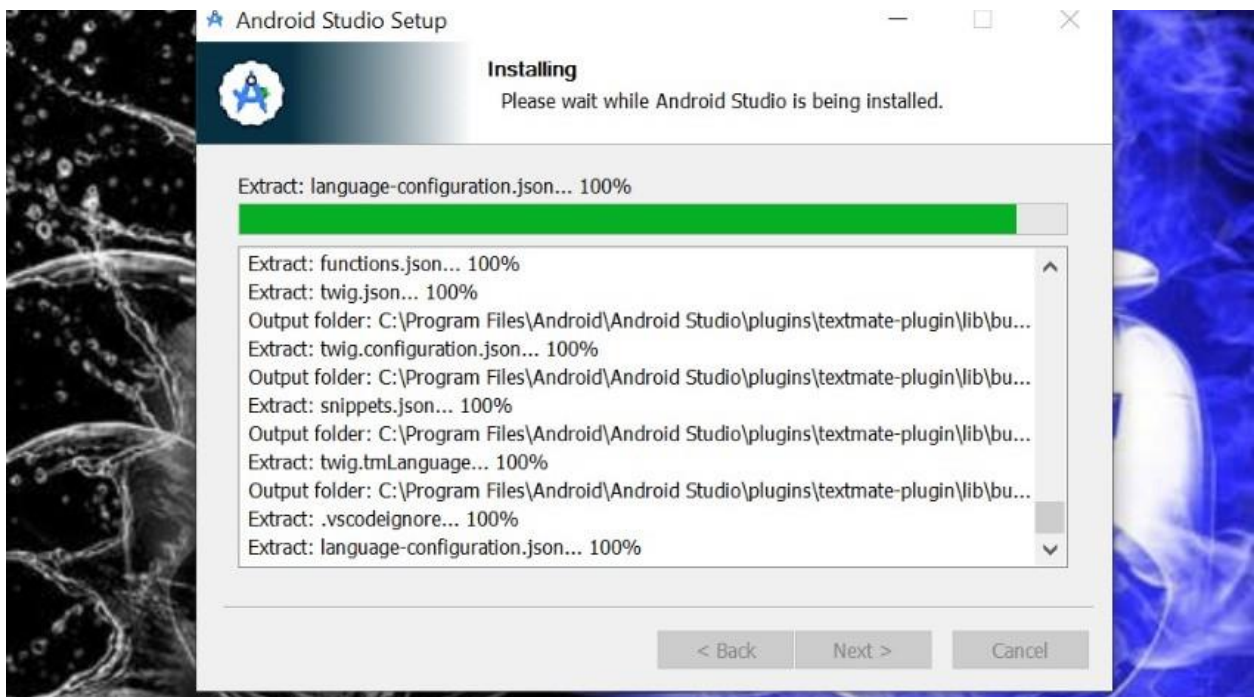
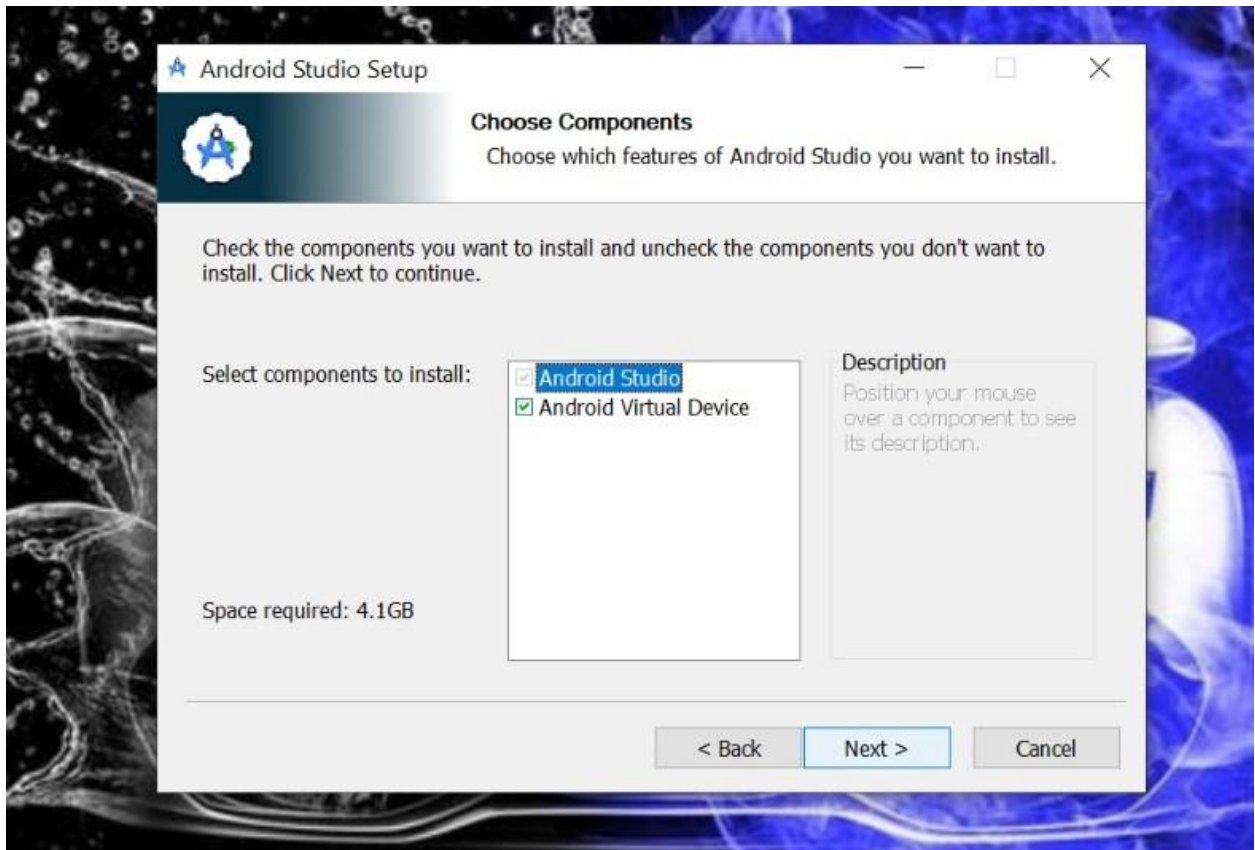
Used Android studio- Kotlin for the assignment and Flutter for the project

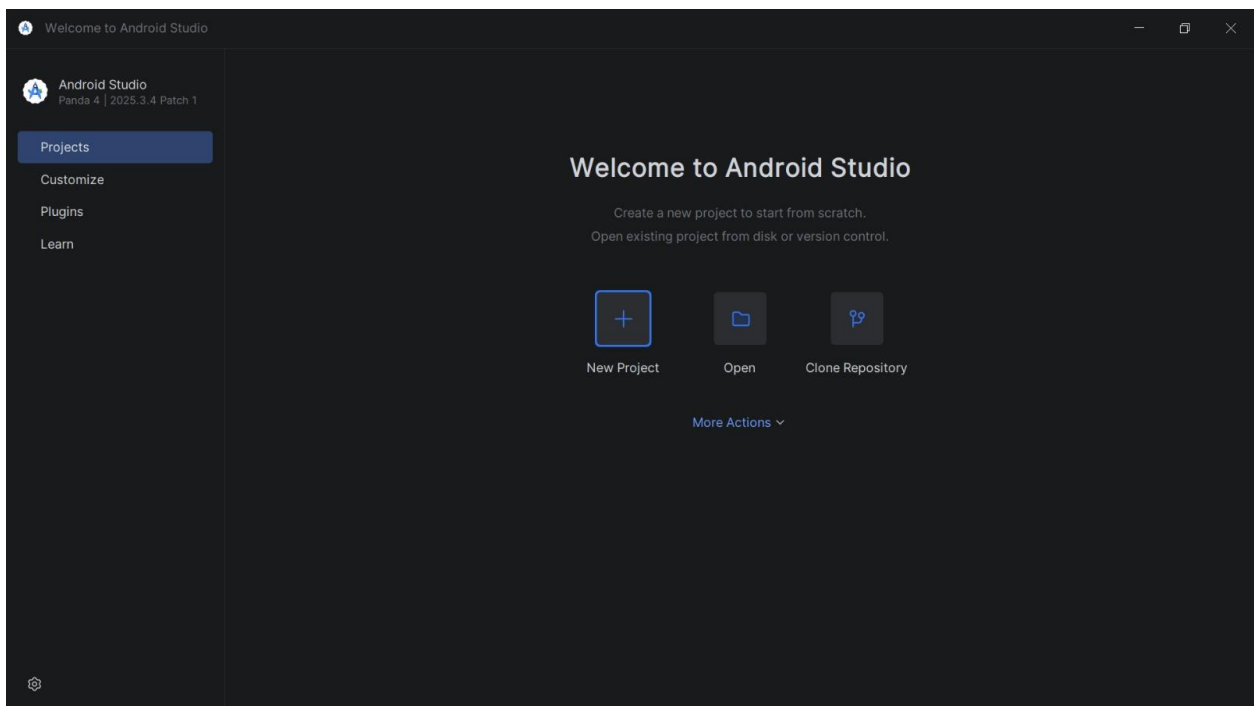
Personal Reflection

Android Studio and Flutter work together in that the IDE provides the heavy duty debugging and SDK engine while Flutter handles the UI magic. Getting them to cooperate on my limited PC was a hectic, RAM freezing nightmare but it was worth it.

Week 2: Mobile Development Environment







Flutter plugin

Programming Language

Flutter

[Google](#) [Plugin homepage](#) ↗

Restart IDE

92.0.0

Overview

What's New

Reviews

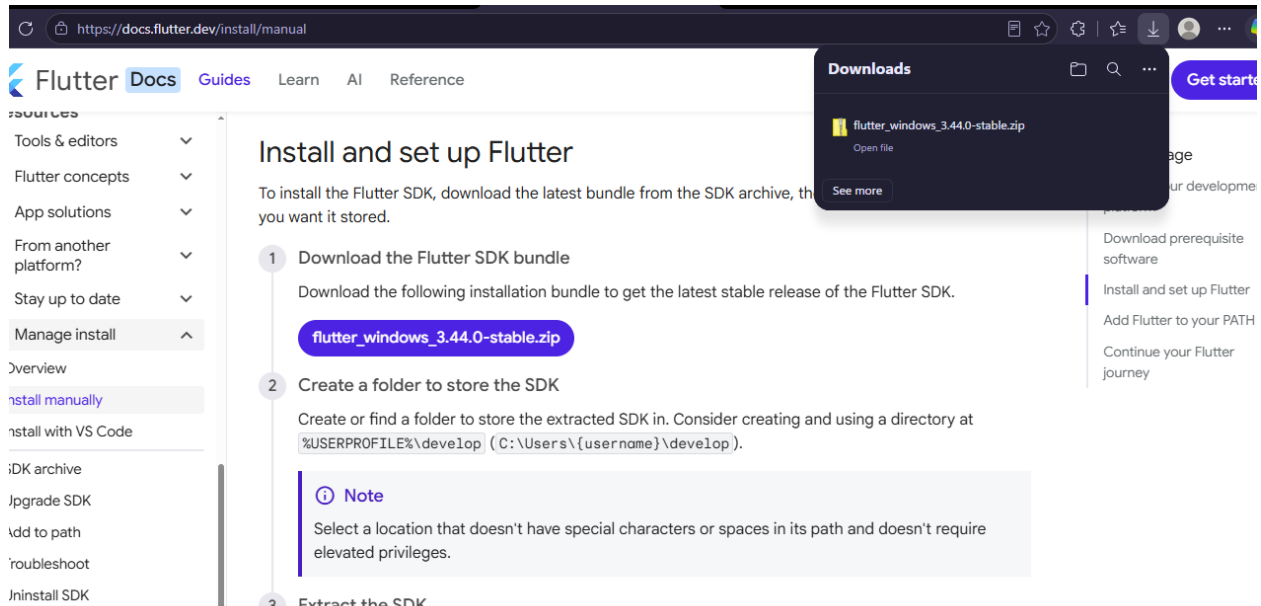
Additional Info

Support for developing Flutter applications. Flutter gives developers an easy and productive way to build and deploy cross-platform, high-performance mobile apps for both Android and iOS. Installing this plugin will also install the Dart plugin.

For some tools, this plugin uses Chromium through JxBrowser to display content from the web. JxBrowser complies with LGPL and offers an option to replace Chromium with another component. To do this:

- Find the JxBrowser files stored in the [plugins directory](#), under `/flutter-intellij/jxbrowser`.
- The LGPL requirements are at [from JxBrowser](#), here you can also download the build script to relink with modified components.

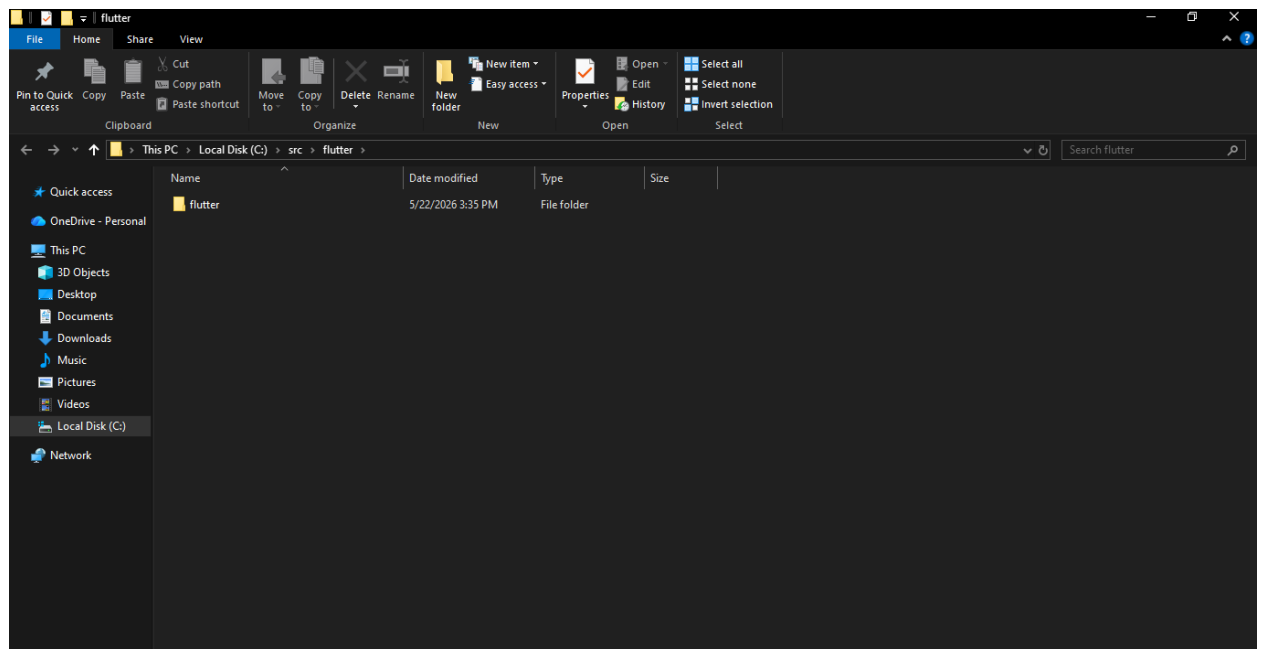
Flutter setup

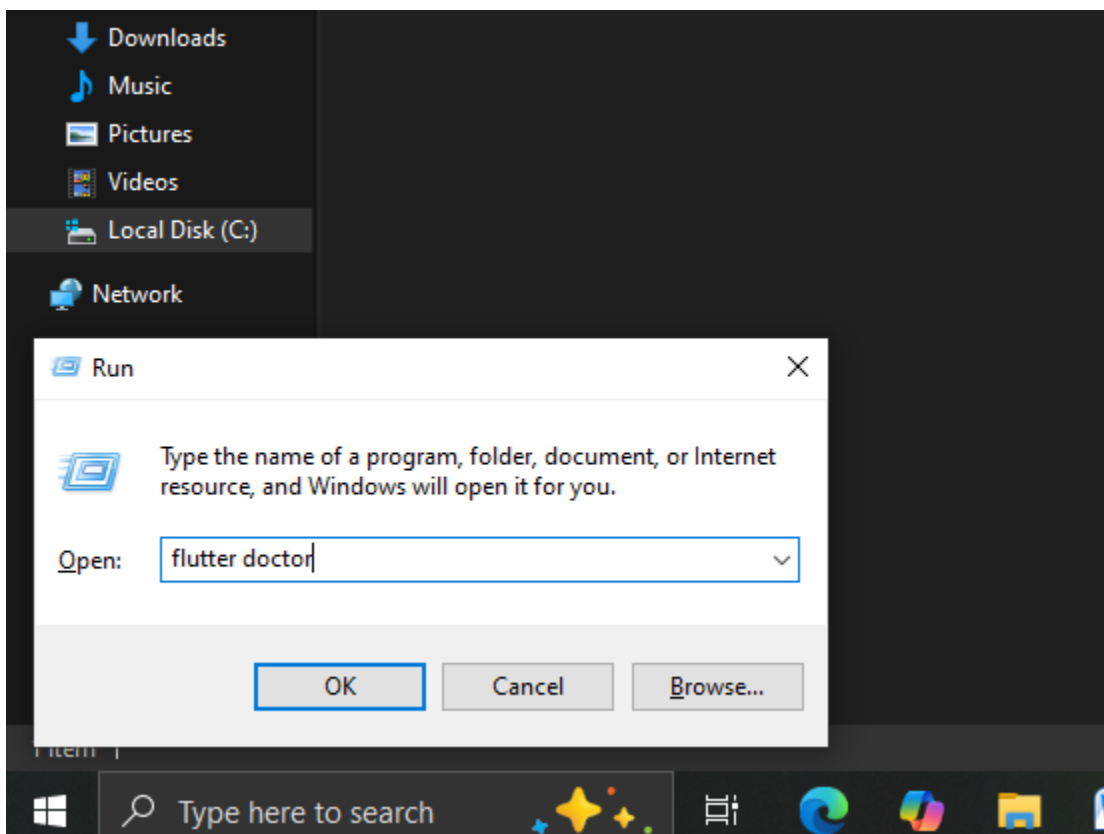
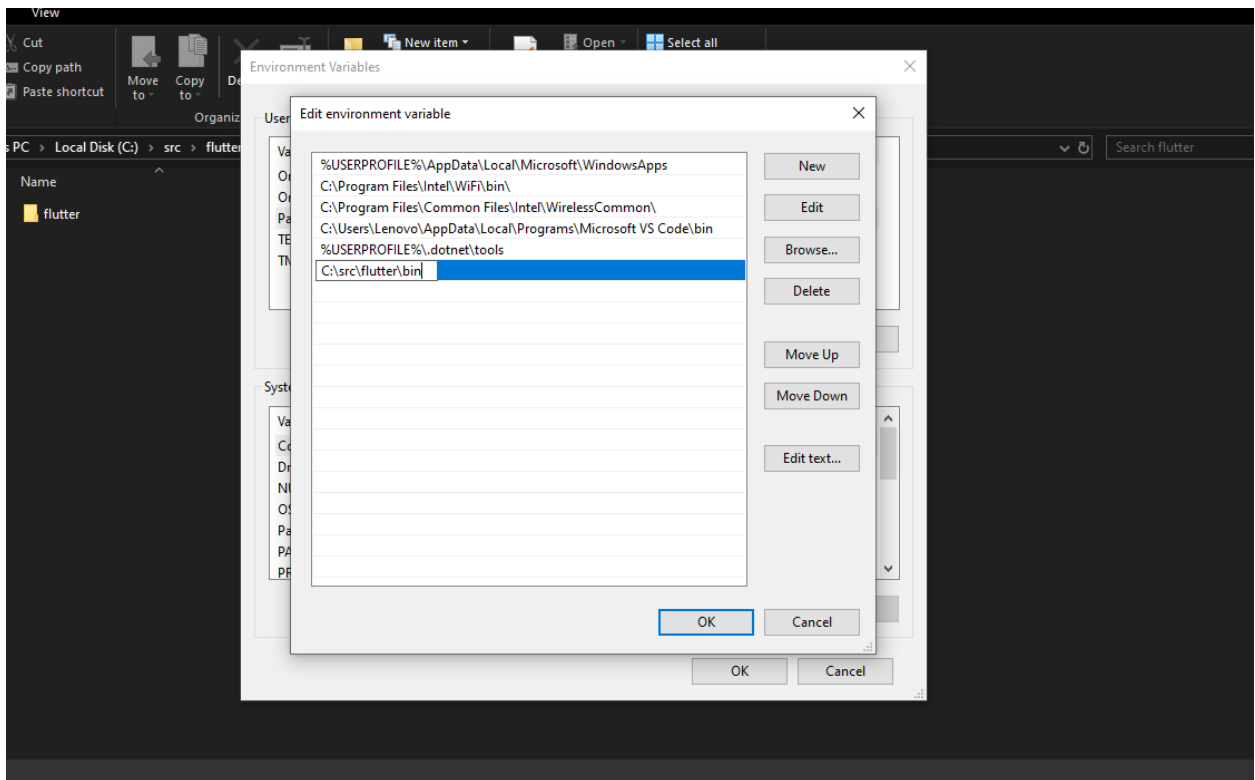


The screenshot shows the Flutter installation manual page at <https://docs.flutter.dev/install/manual>. The page title is "Install and set up Flutter". The instructions are as follows:

- 1 Download the Flutter SDK bundle**
Download the following installation bundle to get the latest stable release of the Flutter SDK.
flutter_windows_3.44.0-stable.zip
- 2 Create a folder to store the SDK**
Create or find a folder to store the extracted SDK in. Consider creating and using a directory at `%USERPROFILE%\develop` (`C:\Users\{username}\develop`).
Note
Select a location that doesn't have special characters or spaces in its path and doesn't require elevated privileges.
- 3 Extract the SDK**

A "Downloads" popup is visible in the top right corner, showing the file **flutter_windows_3.44.0-stable.zip** with an "Open file" button and a "See more" link. The left sidebar of the Flutter docs is also visible, with "Manage install" selected.





```
Command Prompt - flutter doctor
Microsoft Windows [Version 10.0.19045.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>flutter doctor

Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.0, on Microsoft Windows [Version 10.0.19045.5737], locale en-US)
[✓] Windows Version (10 Pro 64-bit, 22H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
    ✗ cmdline-tools component is missing.
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio#command-line-tools-only and make sure
      to set the ANDROID_HOME environment variable.
      See https://developer.android.com/studio/command-line for more details.
    ✗ Android license status unknown.
      Run `flutter doctor --android-licenses` to accept the SDK licenses.
      See https://flutter.dev/to/windows-android-setup for more details.
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.14.32)
    ✗ Visual Studio is missing necessary components. Please re-run the Visual Studio installer for the "Desktop
      development with C++" workload, and include these components:
        MSVC v142 - VS 2019 C++ x64/x86 build tools
        - If there are multiple build tool versions available, install the latest
        C++ CMake tools for Windows
        Windows 10 SDK
[✓] Connected device (3 available)
[✓] Network resources

! Doctor found issues in 2 categories.

C:\Users\Lenovo>
```

Appearance & Behavior

Appearance

Menus and Toolbars

System Settings

File Colors

Scopes

Notifications

Data Editor and Viewer

Quick Lists

Required Plugins

Trusted Locations

Path Variables

Presentation Assistant

Keymap

Editor

Plugins

Version Control

Build, Execution, Deployment

Languages & Frameworks

C/C++

Android SDK

JVM Logging

Kotlin

Languages & Frameworks > Android SDK Revert changes

Manager for the Android SDK and Tools used by the IDE

Android SDK Location: C:\Users\Lenovo\AppData\Local\Android\Sdk Edit Optimize disk space

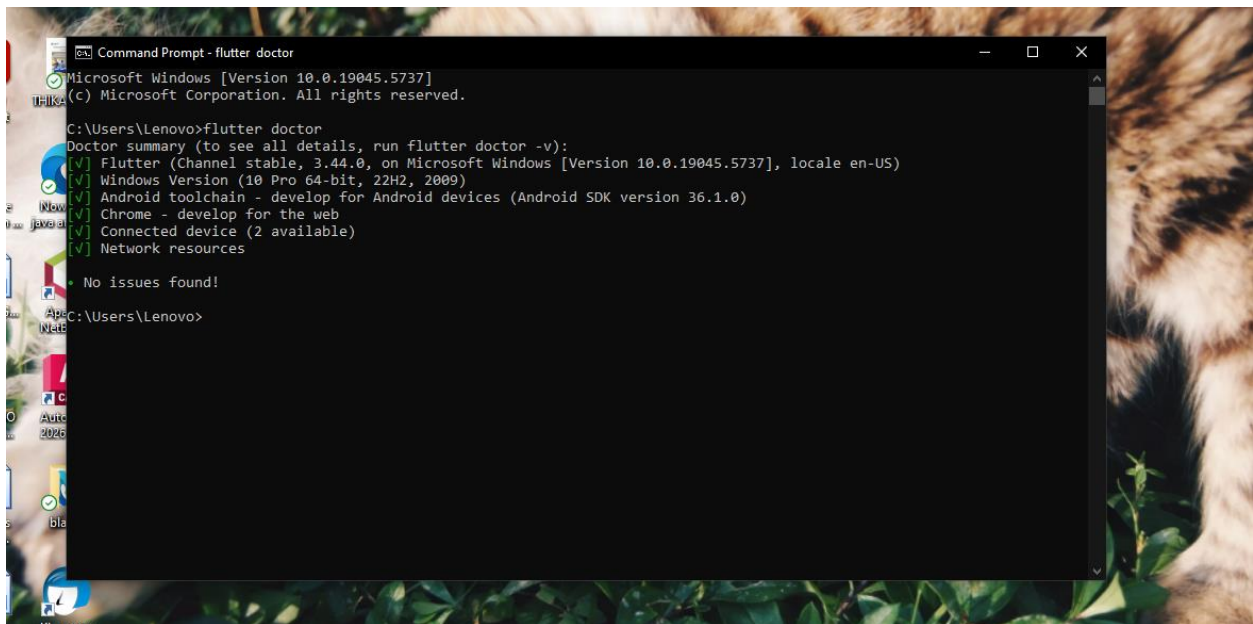
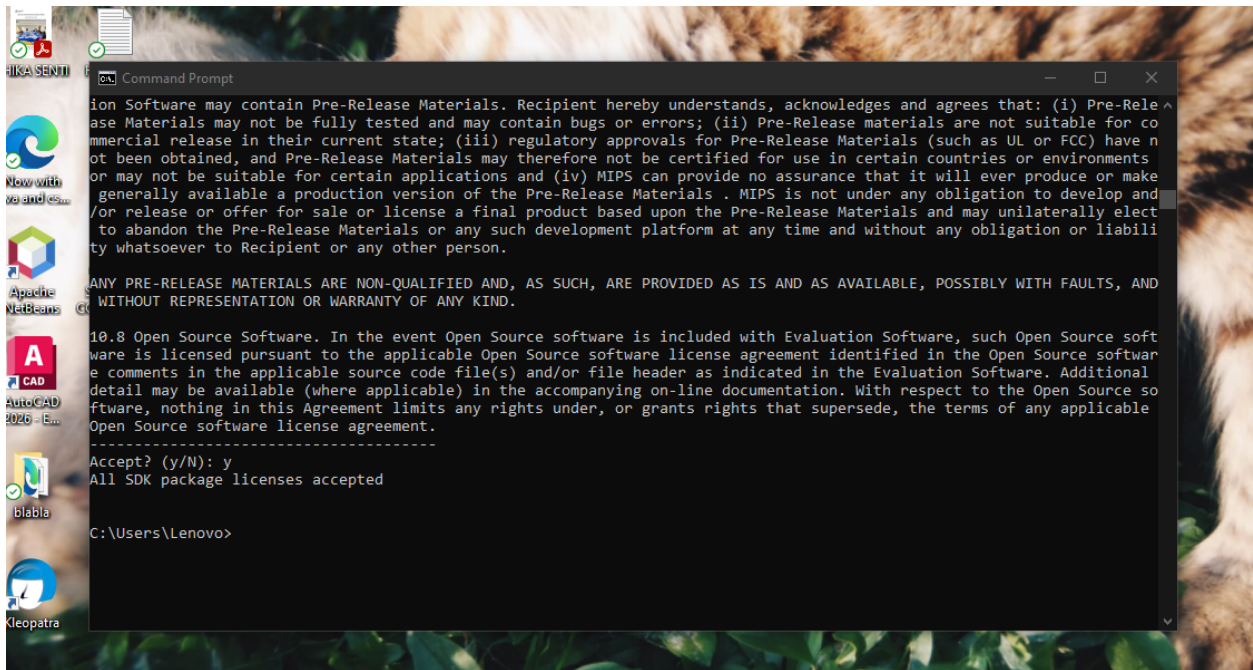
SDK Platforms SDK Tools SDK Update Sites

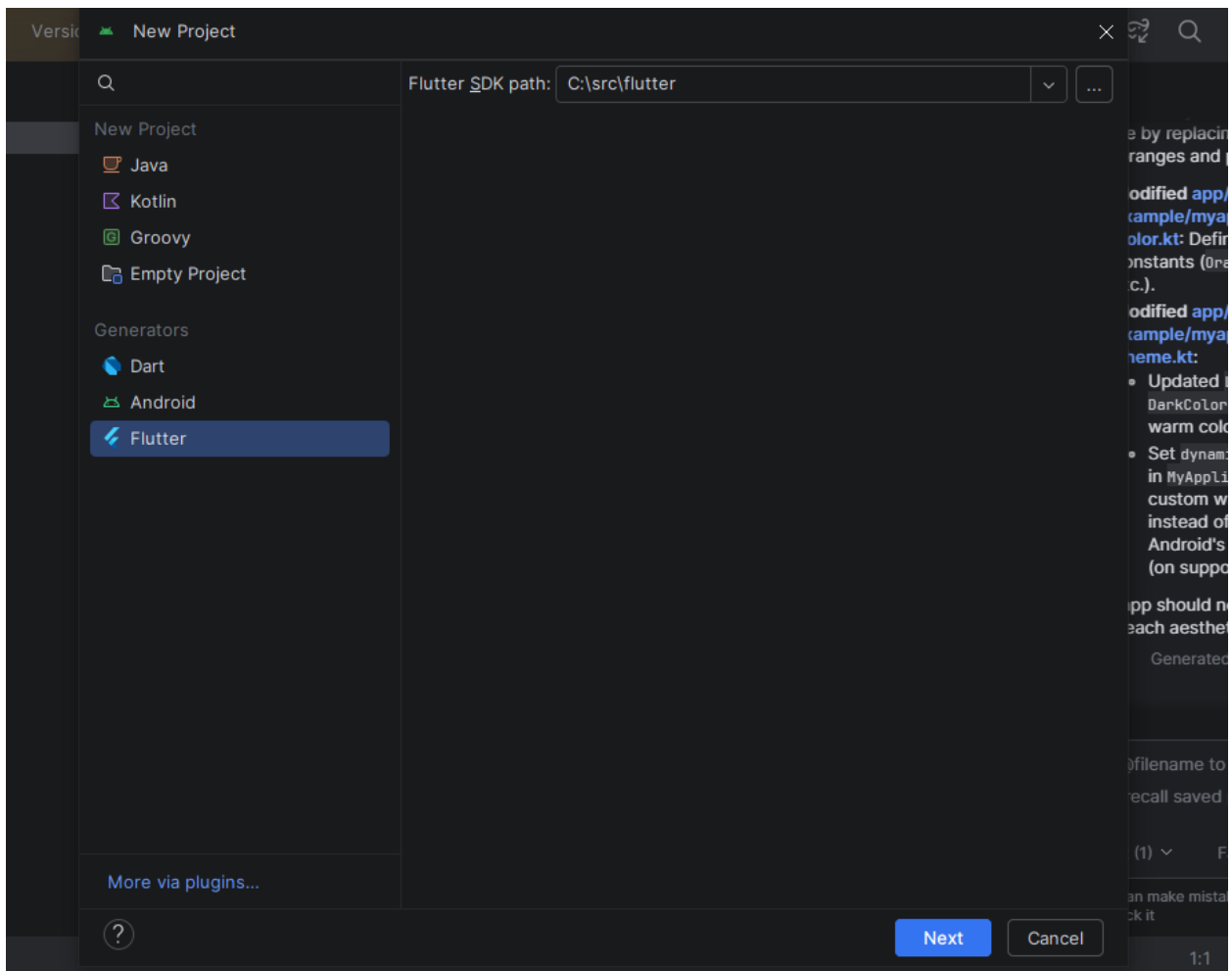
Below are the available SDK developer tools. Once installed, the IDE will automatically check for updates. Check "show package details" to display available versions of an SDK Tool.

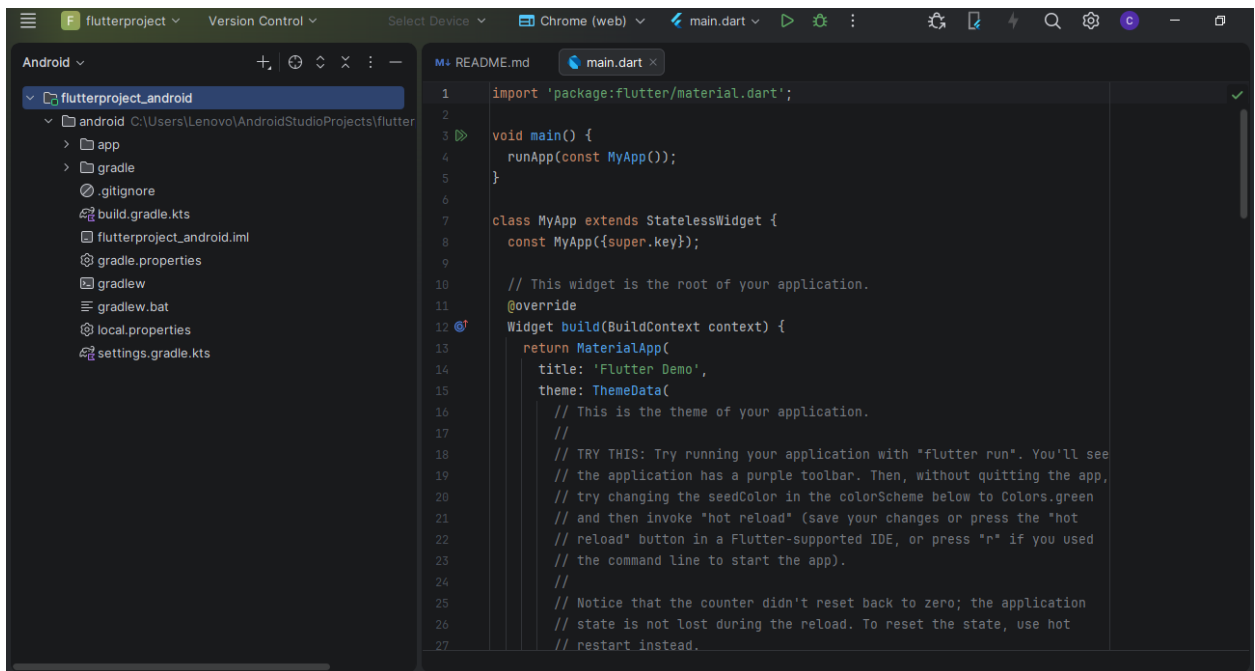
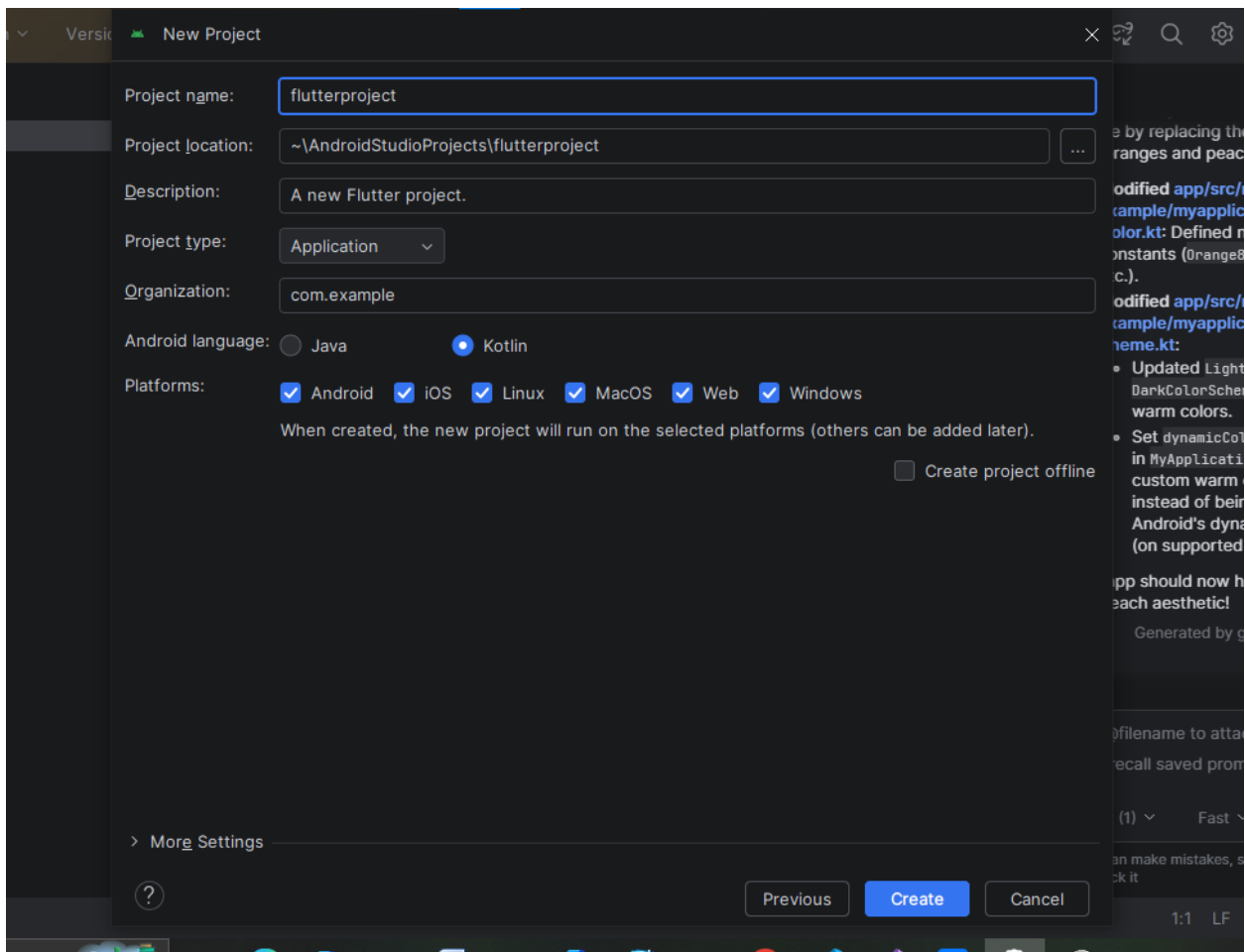
Name	Version	Status
☒ Android SDK Build-Tools		Installed
☐ NDK (Side by side)		Not Installed
☒ Android SDK Command-line Tools (latest)		Installed
☐ CMake		Not Installed
☐ Android Auto API Simulators	1	Not installed
☐ Android Auto Desktop Head Unit Emulator	2.0	Not installed
☒ Android Emulator	36.5.11	Installed
☒ Android Emulator hypervisor driver (installer)	2.2.0	Installed
☐ Android Project Templates	0.1.0	Not installed
☒ Android SDK Platform-Tools	37.0.0	Installed
☐ Android Support Repository	47.0.0	Not installed
☐ Google Play APK Expansion library	1	Not installed
☐ Google Play Instant Development SDK (Deprecated)	1.9.0	Not installed
☐ Google Play Licensing Library	1	Not installed

☒ Hide Obsolete Packages ☐ Show Package Details by editing and lines of code... not show again

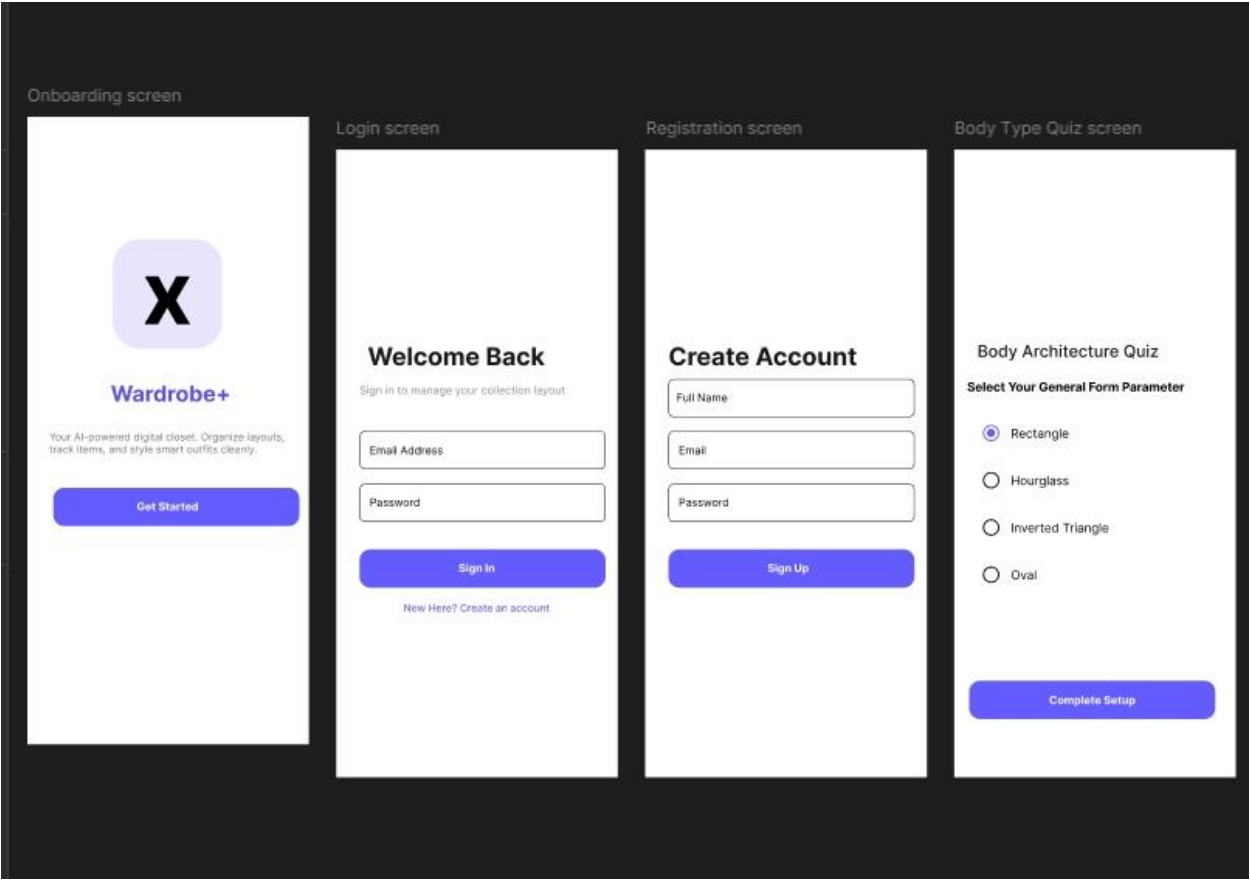
OK Cancel Apply



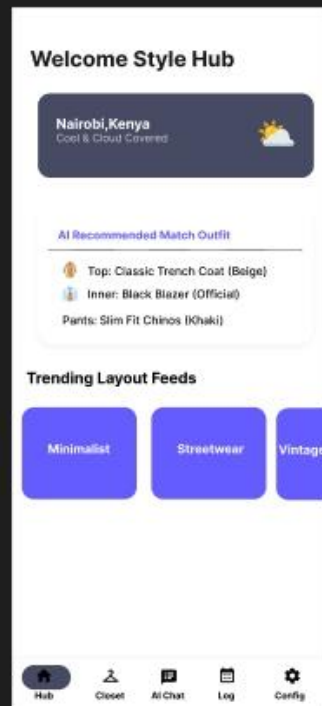




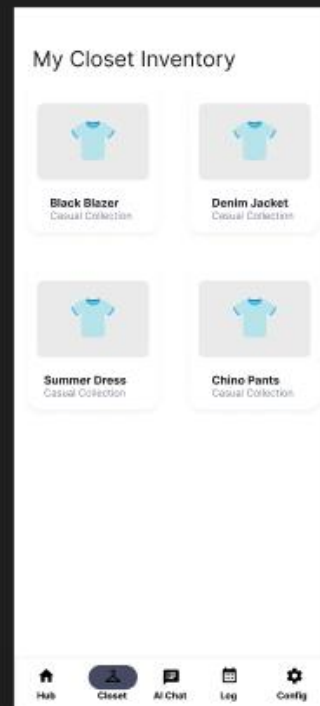
Week 3: User Interface Design



Dashboard

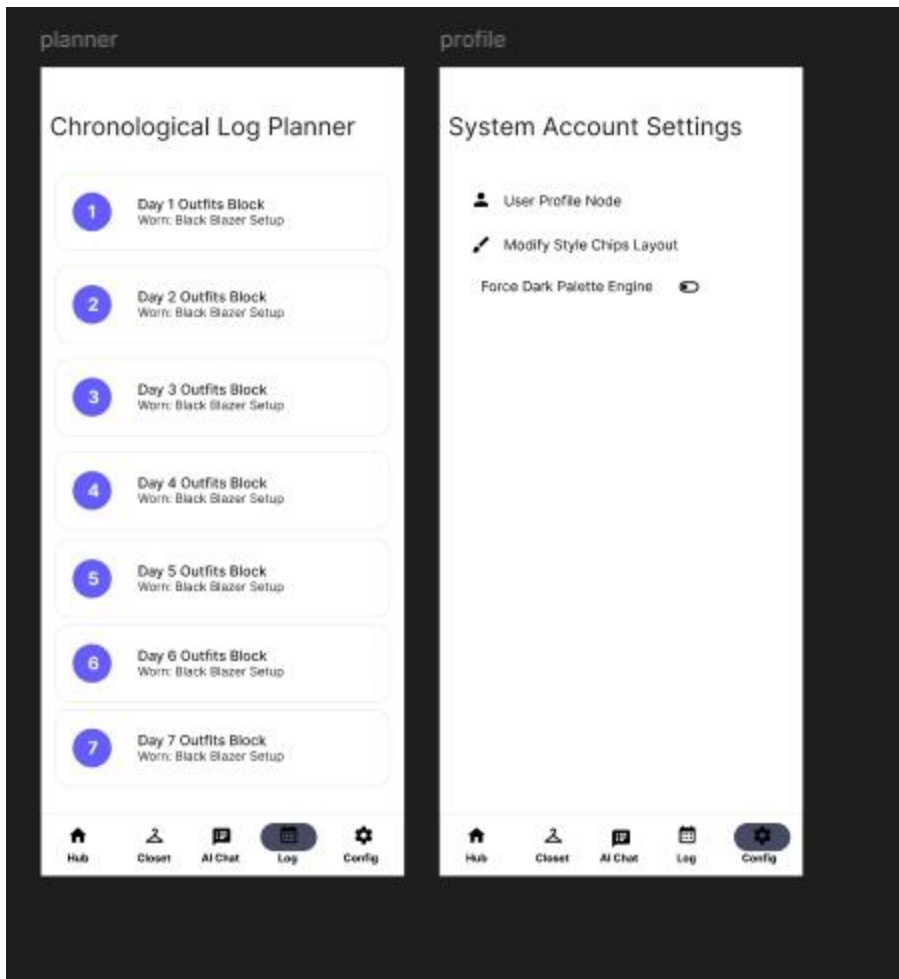


Wardrobe screen



chat screen





Tools used: Figma

Week 4: Event Handling and User Interaction

Figure 1

Figure 2

Figure 3

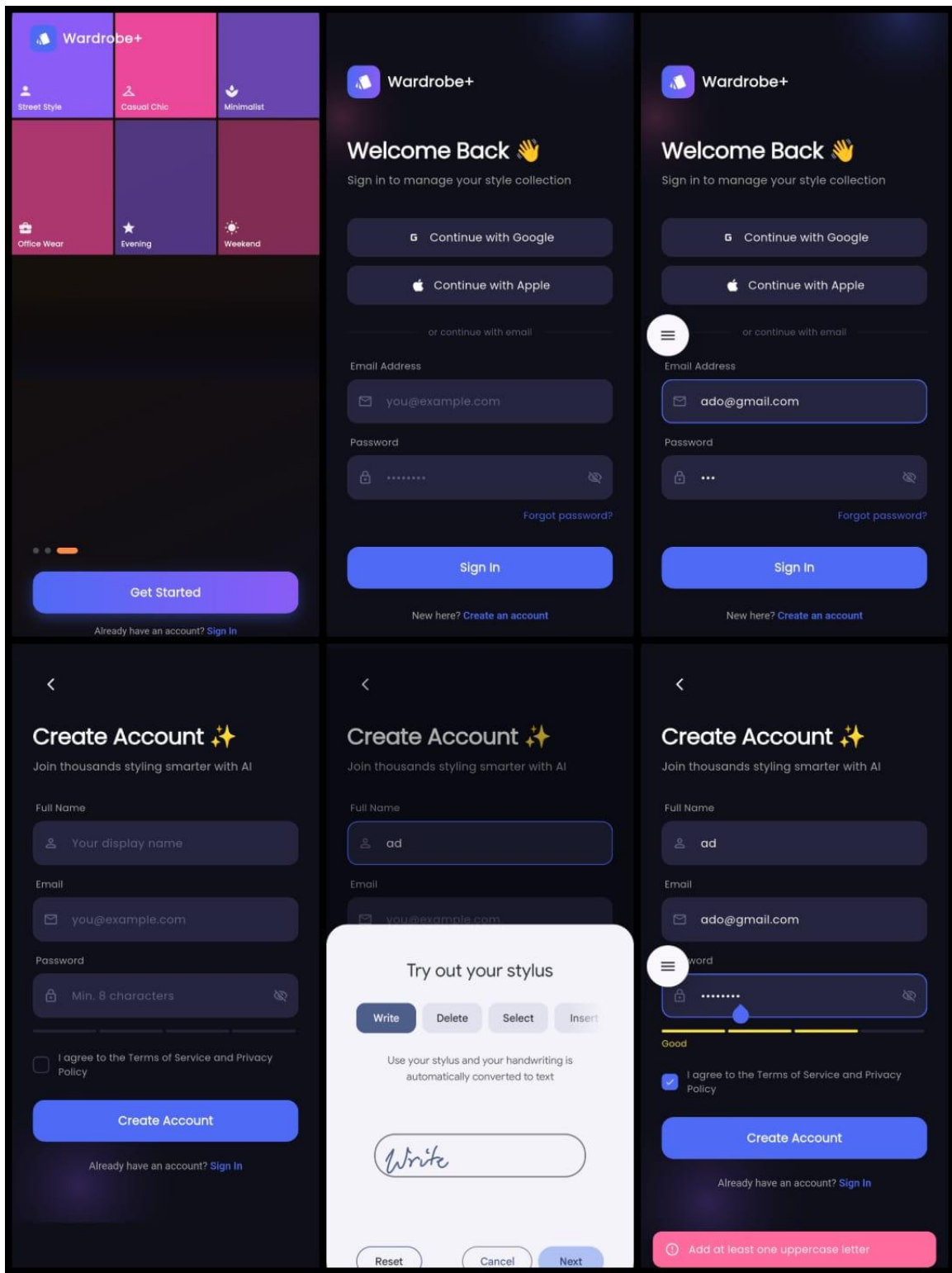


Figure 4

Figure 5

Figure 6

Figure 1: Onboarding screen

Figure2 &3: Login Screen

Figure 4, 5&6: Registration screen

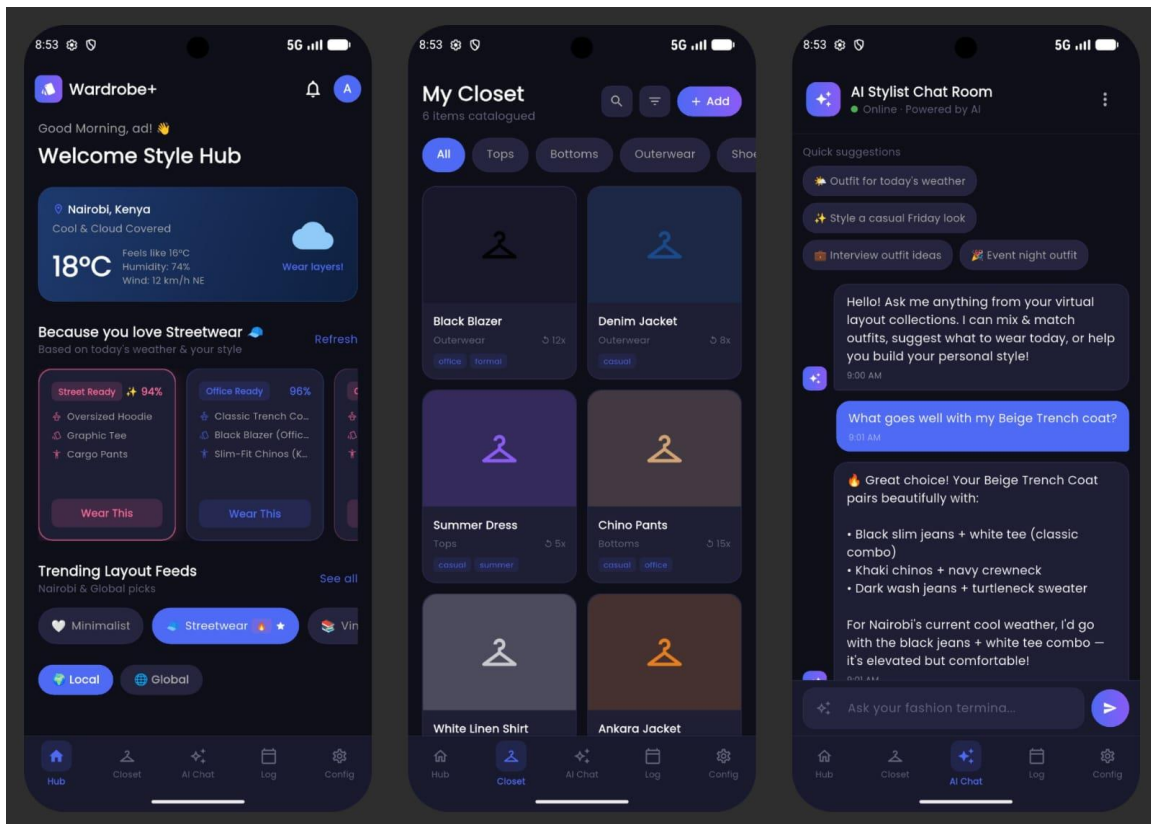


Figure 7

Figure 8

Figure 9

Figure 7:dashboard_screen

Figure 8: closet_screen

Figure 9: ai_chat-screen

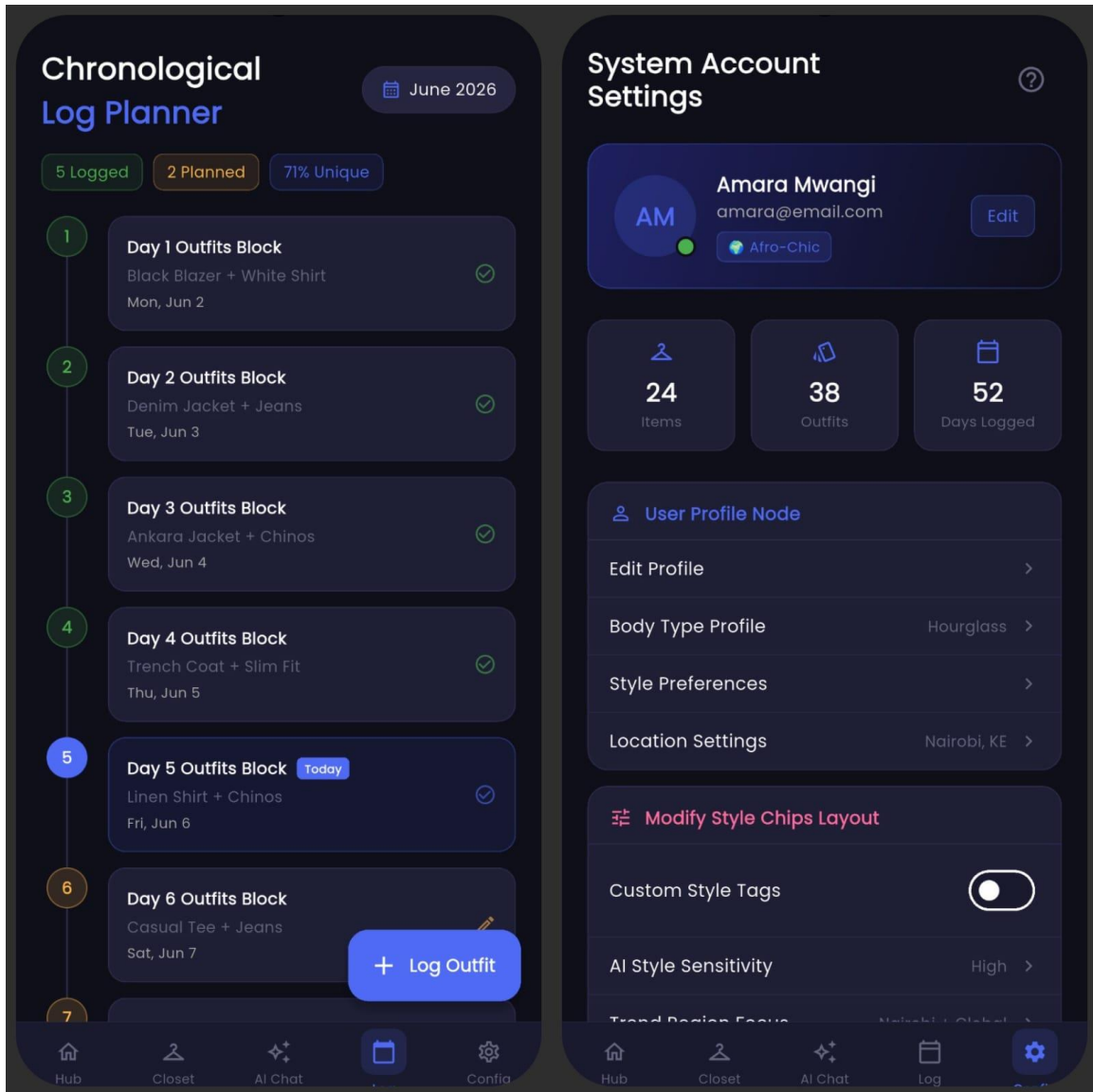


Figure 10

Figure 11

Figure 10:planner_screen

Figure 11:user_profile_screen

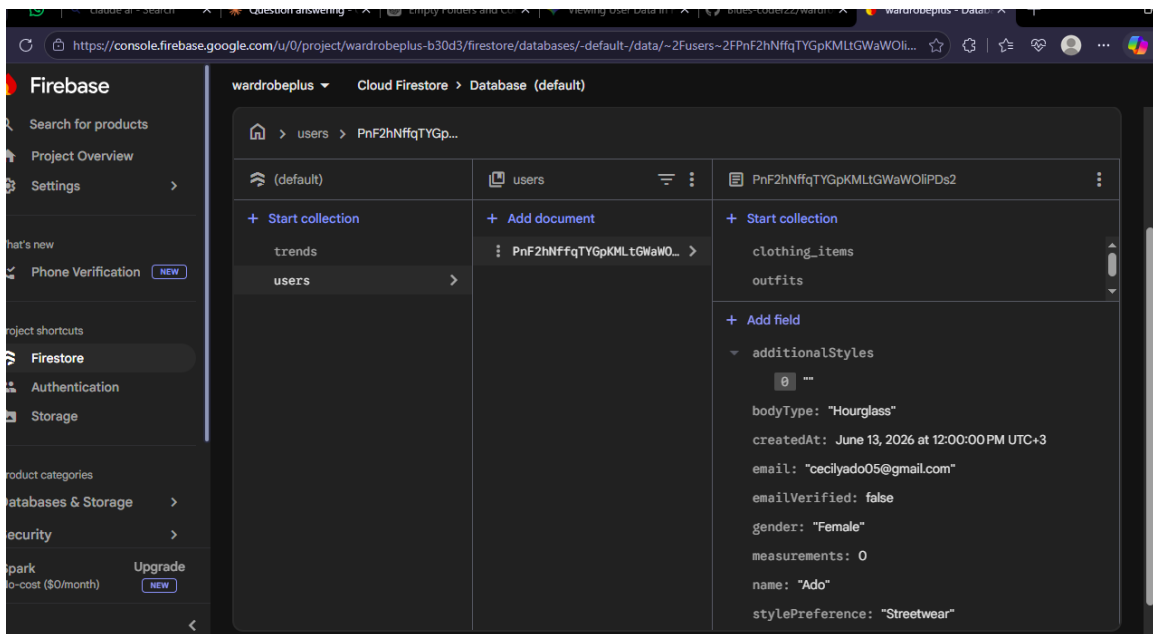
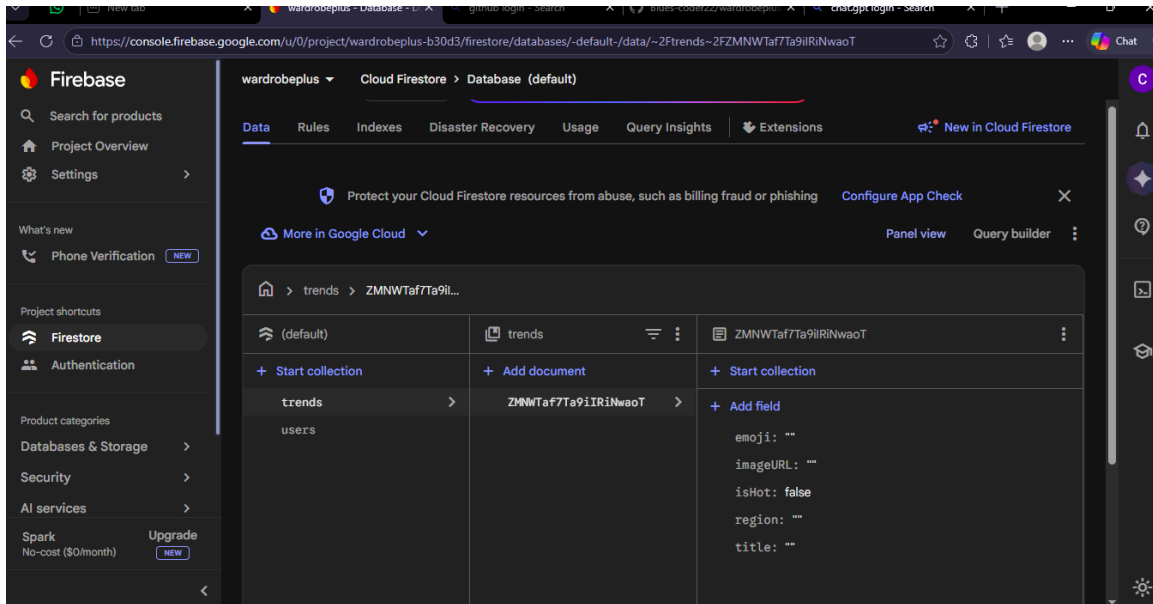
Tools used: Flutter, Android studio, Dart

Personal Reflection: Learned how user actions trigger events and how Flutter widgets respond to user input. Understanding navigation routes improved the app's usability.

Week 5: Data Management

Database structure

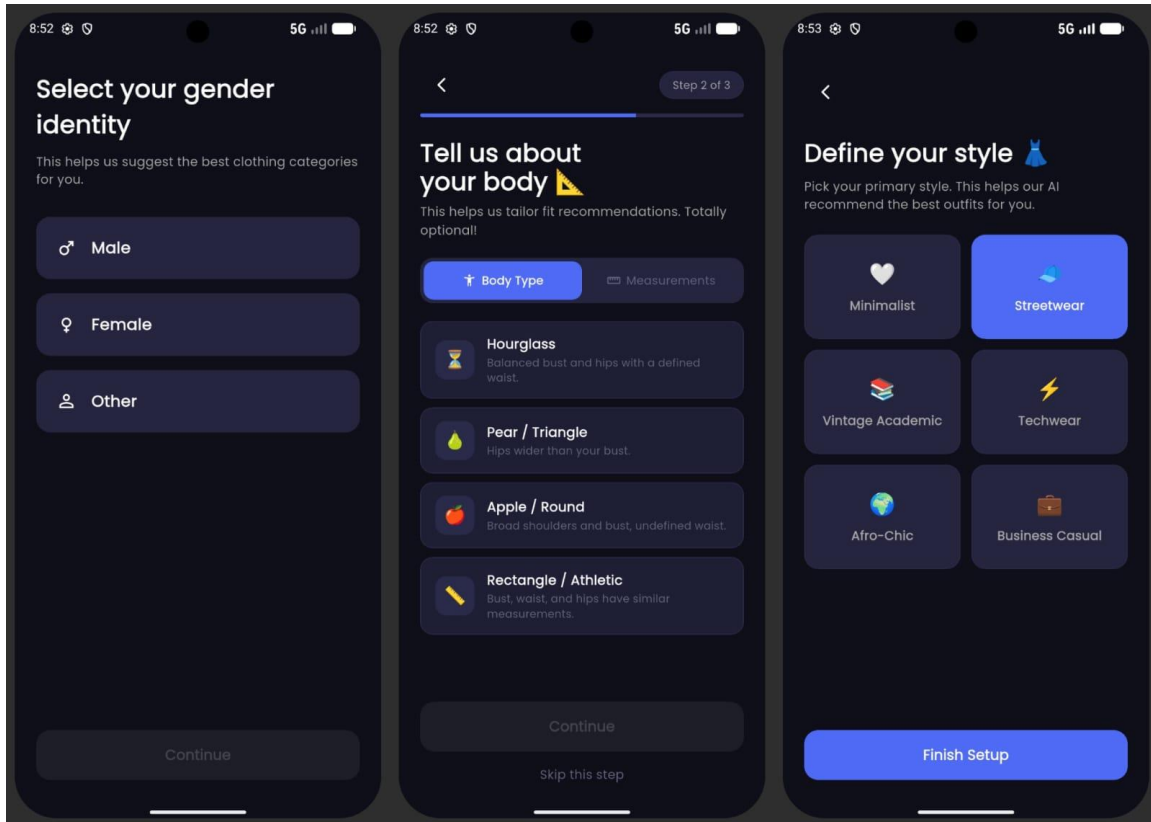
Uses NoSQL (Cloud Firestore) was chosen because it is scalable, flexible, supports real-time updates, works well with Flutter, and stores data as documents instead of complex relational tables.



CRUD Operation

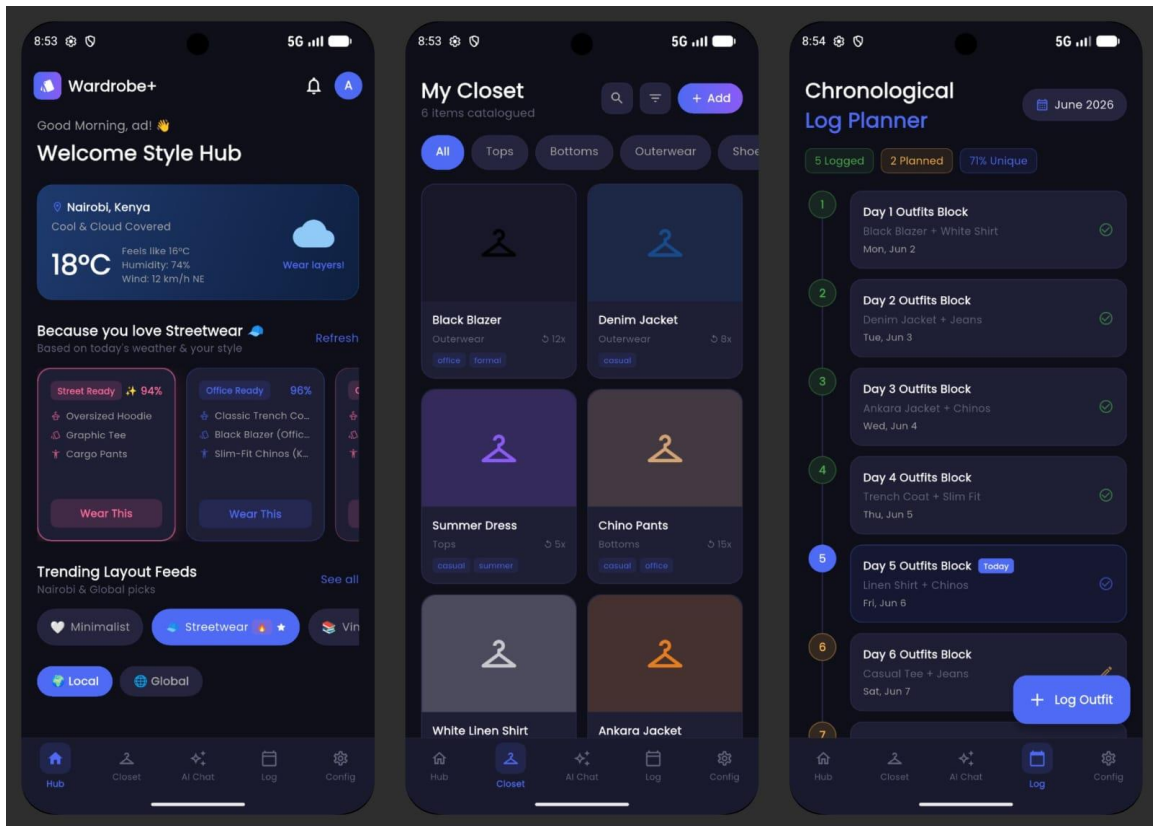
CREATE (Register user)

```
await _usersCollection.doc(fbUser.uid).set(user.toMap());
```



READ (Fetch user details)

```
final snapshot = await docRef.get();
```



UPDATE (Profile Update)

```
await _usersCollection.doc(_currentUser!.id).set(
  updates,
  SetOptions(merge: true),
);
```




24

Items



38

Outfits



52

Days Logged

User Profile Node

Edit Profile >

Body Type Profile Hourglass >

Style Preferences >

Location Settings Nairobi, KE >

Modify Style Chips Layout

Custom Style Tags ☒

AI Style Sensitivity High >

Trend Region Focus Nairobi, Global >



Hub



Closet



AI Chat



Log

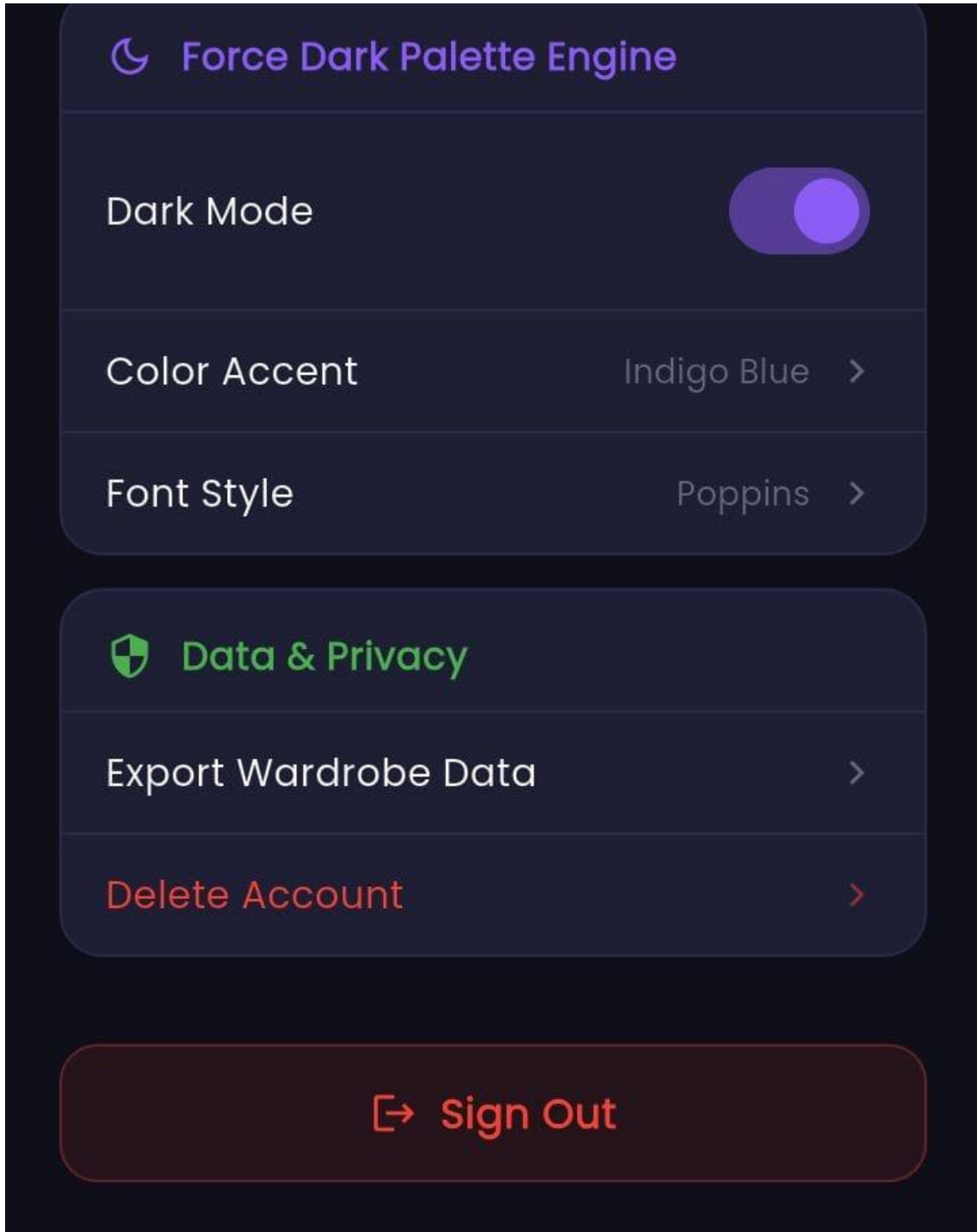


Config

DELETE (Sign Out cache clear)

```
await _auth.signOut();
```

```
await LocalStorageService.clearUserData();
```



Authentication Integration

Firebase authentication manages user login and registration. Passwords are not stored in Firestore, improving security.

8:51



5G



Create Account ✨

Join thousands styling smarter with AI

Full Name

ad

Email

ado@gmail.com

word



.....



Good



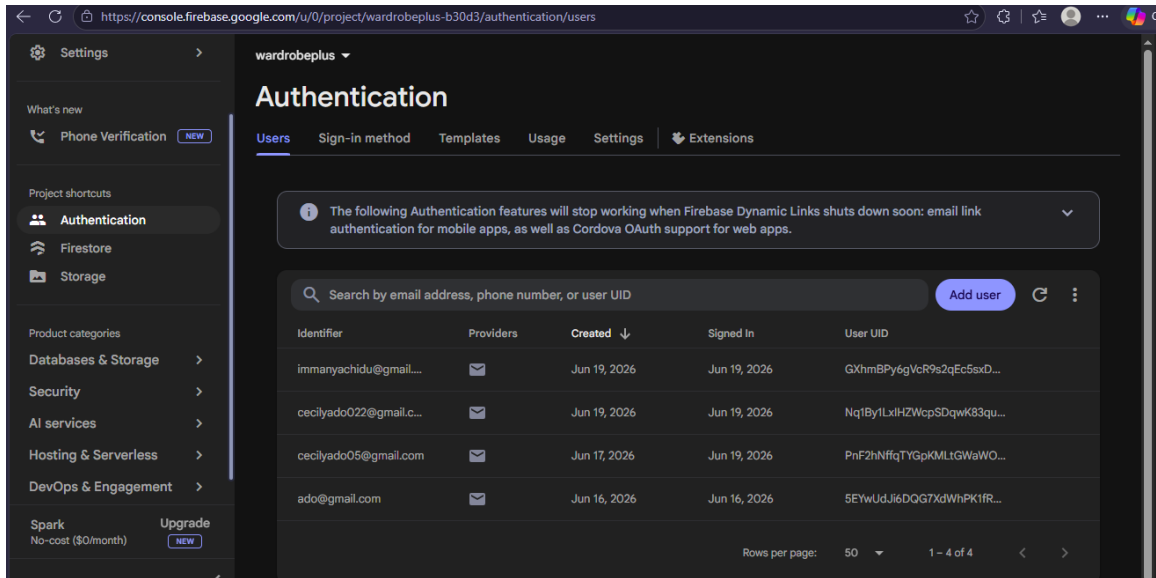
I agree to the Terms of Service and Privacy Policy

Create Account

Already have an account? [Sign In](#)



Add at least one uppercase letter



Architecture Explanation

Flutter App



AuthService



Firebase Auth (Identity)



Cloud Firestore (users collection)



LocalStorageService (cache)

Tools/Software Used:

Flutter (Dart)

Firebase Firestore

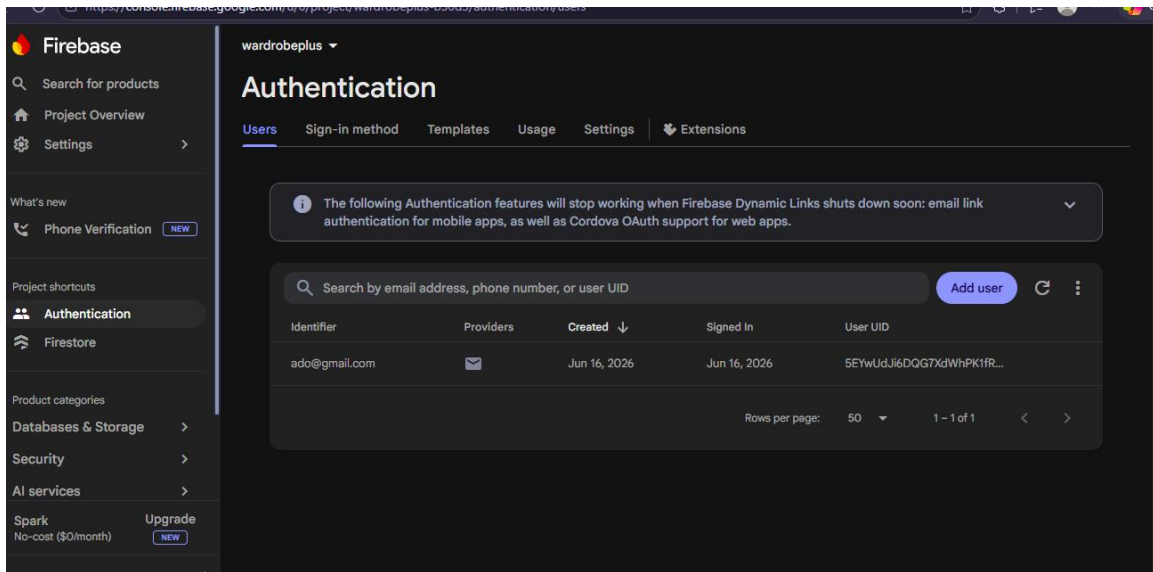
Firebase Authentication

Android Studio

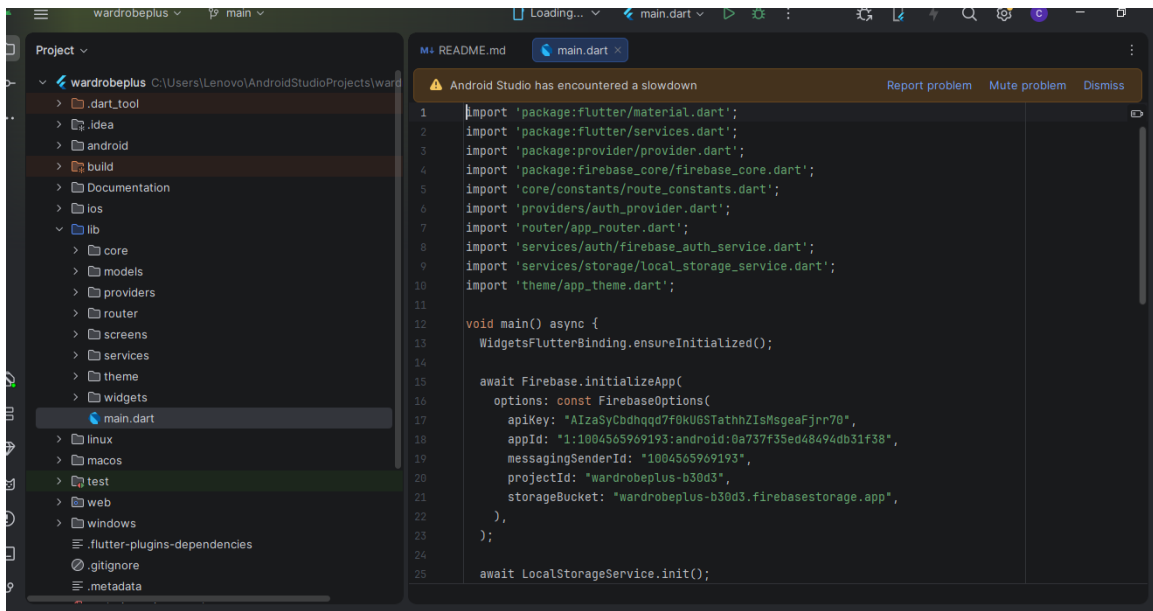
Local Storage (Shared preferences)

Personal Reflection: Gained experience storing and retrieving data from the cloud, making the application more dynamic.

Week 6: Networking and APIs



Flutter-Firebase API



Why Networking:

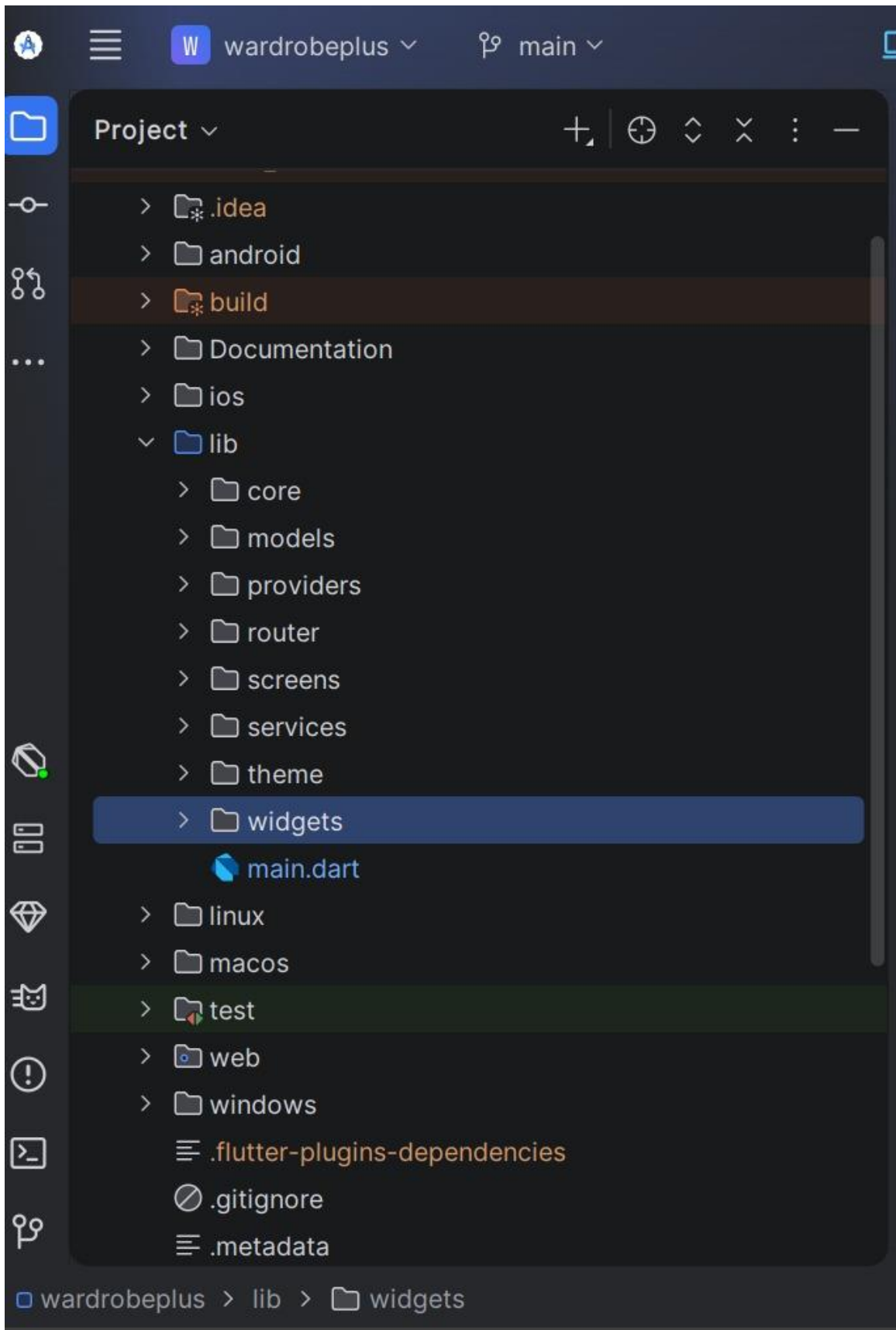
Networking is important because it allows the app to communicate with online services such as Firebase Authentication and Firestore. Through networking, users can create

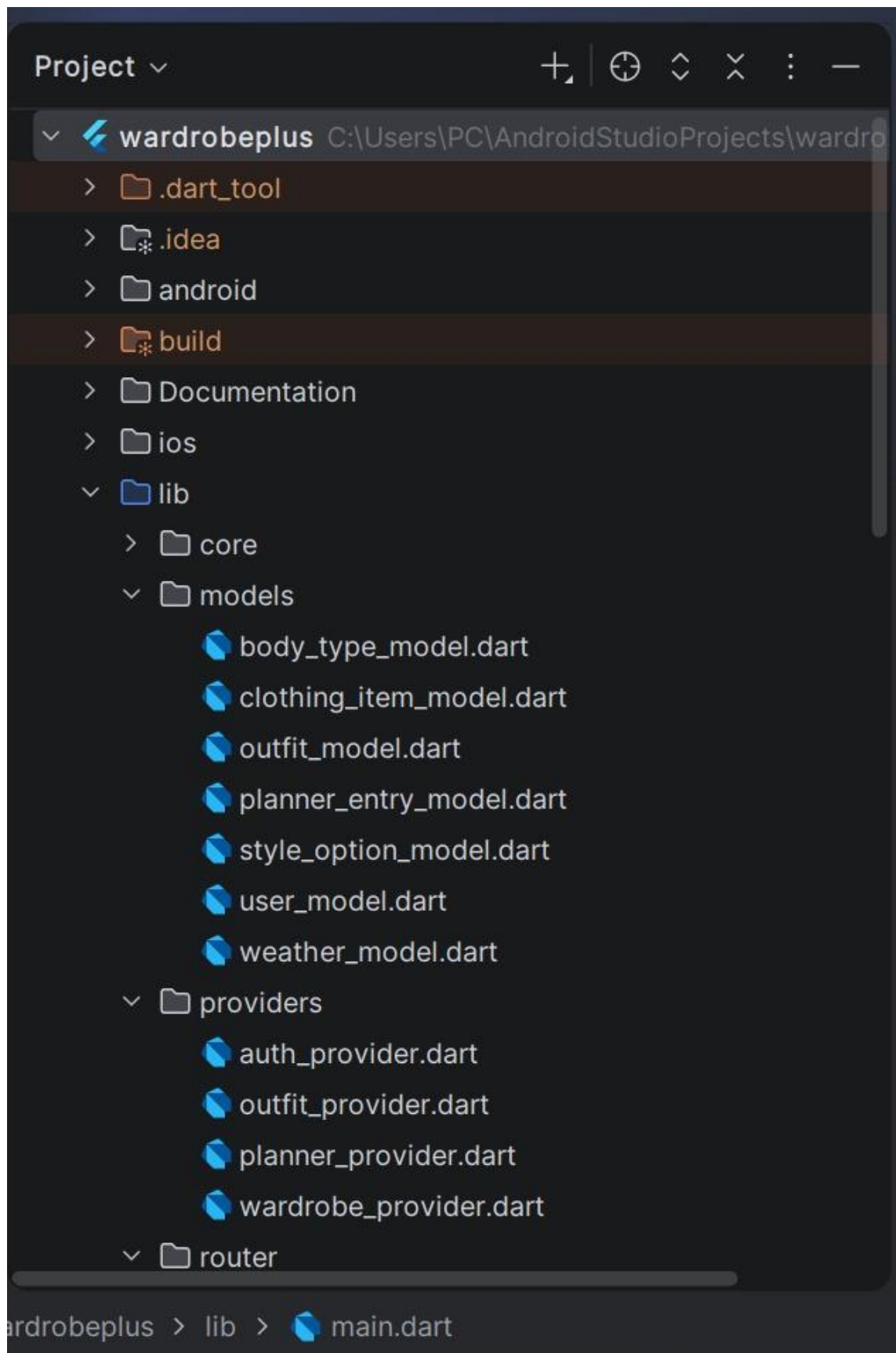
accounts, log in, store their preferences, and access their data from different devices. It also enables future features such as AI recommendations and cloud image storage. Without networking, the app would be limited to storing data only on the user's device.

Tools/Software Used: Firebase, Flutter, Android Studio

Personal Reflection: Learned how mobile applications communicate with backend services and how cloud databases improve accessibility.

Week 7: Application Architecture





Project ▾



- planner_provider.dart
- wardrobe_provider.dart

▾ router

- app_router.dart
- auth_guard.dart

▾ screens

- > auth
- > main
- > onboarding
- > premium
- > splash

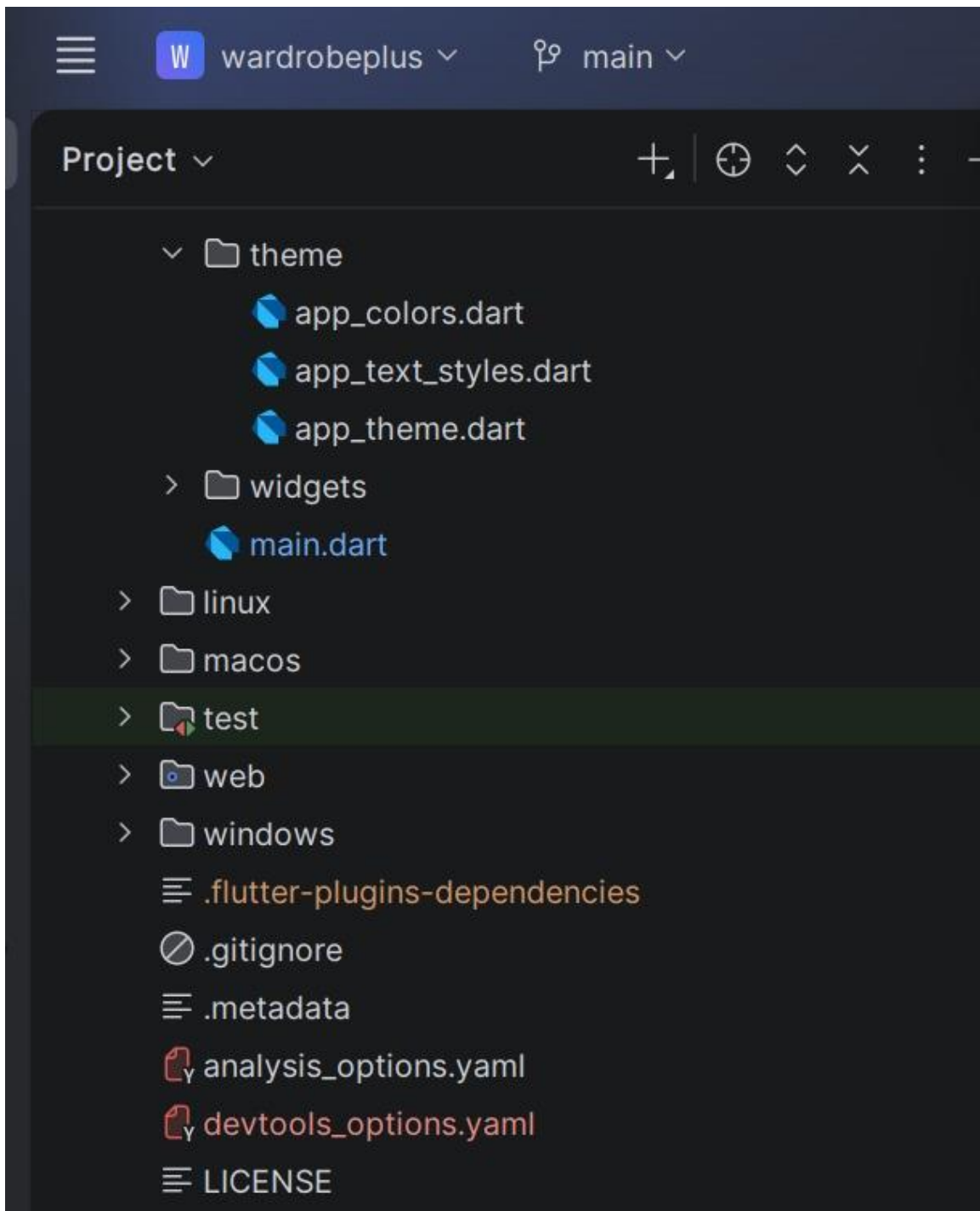
▾ services

- > ai
- > auth
- ▾ firestore
 - firestore_service.dart
- > recommendation
- > storage
- > trends
- > weather

▾ theme

- app_colors.dart
- app_text_styles.dart

wardrobeplus > lib > main.dart



Week 8: Project Planning

Week 9: Device Features and Sensors

Same as week 1

Week 10: Notifications and Background Services

Same as week 1

Week 11: Mobile Security

Same as week 1

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic

Major Challenges Faced

How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: _____

Student Signature: _____

Date: _____

Lecturer Name: _____

Lecturer Remarks: _____

Date: _____